
AI-Assisted Coding for Educators

AI-assisted coding

AI coding tools

Human thinking + AI collaboration

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- Quantum Computing
- Bio Science
- Music Innovation
- Video Arts
- Work-based learning
- Cybersecurity
- Robotics
- Game development
- AI
- ...



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Agenda

- Generative AI
- AI assisted Coding
 - Tools
 - Risks
- Takeaways



GenAI

Artificial Intelligence

Machine Learning

Deep Learning

Generative AI



Artificial Intelligence

Intelligence demonstrated by machines



Machine Learning

Learn from data



Deep Learning

Model after the human brain (Neural Networks)



Generative AI

Create new written, visual, and auditory content

Generative AI

The best thing about AI is its ability to ...

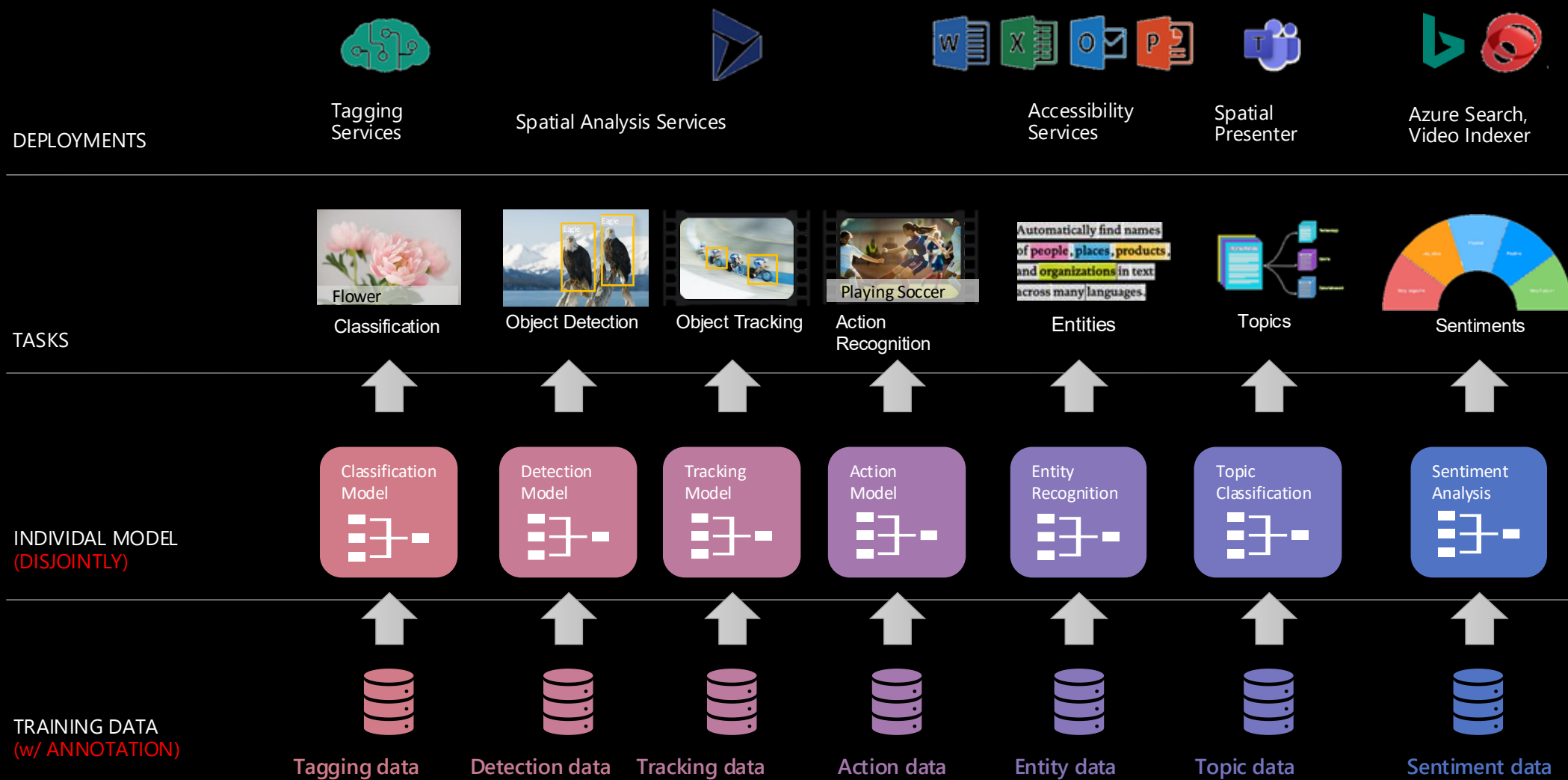
Adapt = 30%

Process = 22%

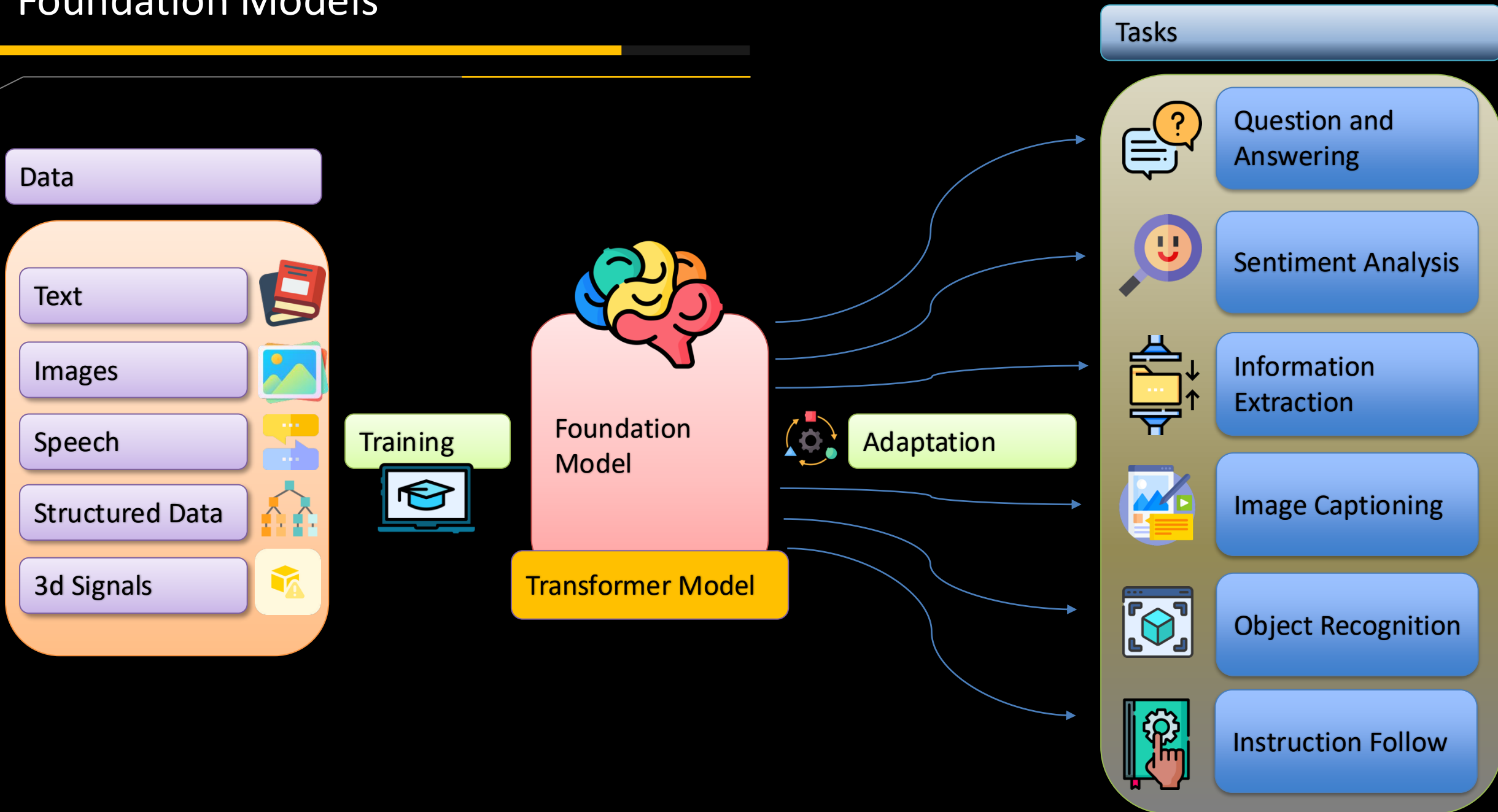
Analyze = 7%

Traditional model development

High cost & slow deployment - Each service is trained disjointly



Foundation Models

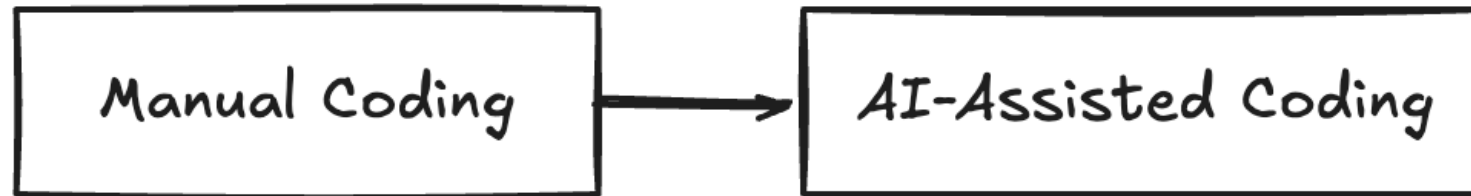


Why This Matters

- AI is changing software development
- Students will work with AI professionally
- Goal: teach thinking, not memorization



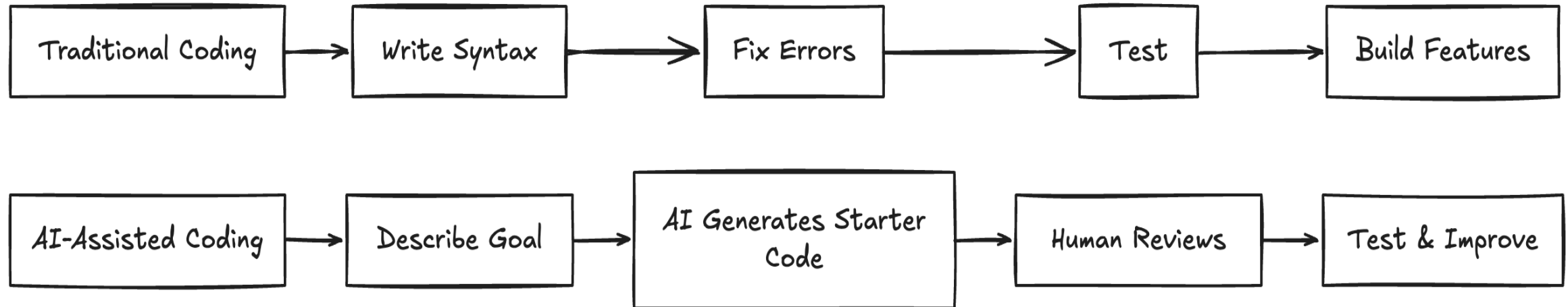
Calculator Analogy



- Calculators changed math education
- AI coding tools change (programming) education
- Humans still provide reasoning and verification

- Quantum Computing
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- ...

Traditional vs AI-Assisted Coding

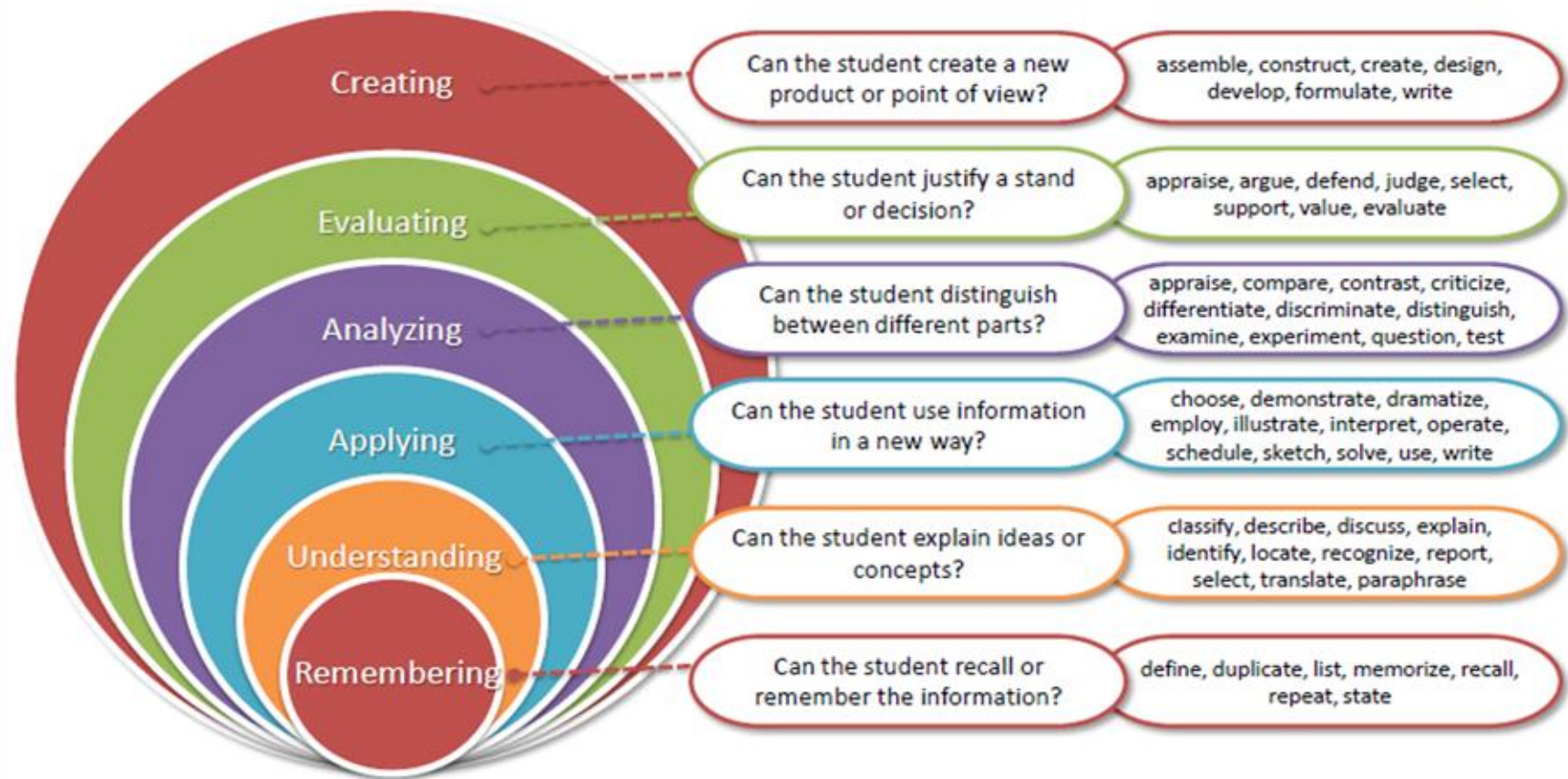


- Before AI: syntax, debugging, boilerplate
- With AI: design, testing, evaluation
- Shift toward higher-order thinking

Design

- Learning objectives
- Select instructional strategies
- Select assessment methods

Bloom's Taxonomy (Revised)

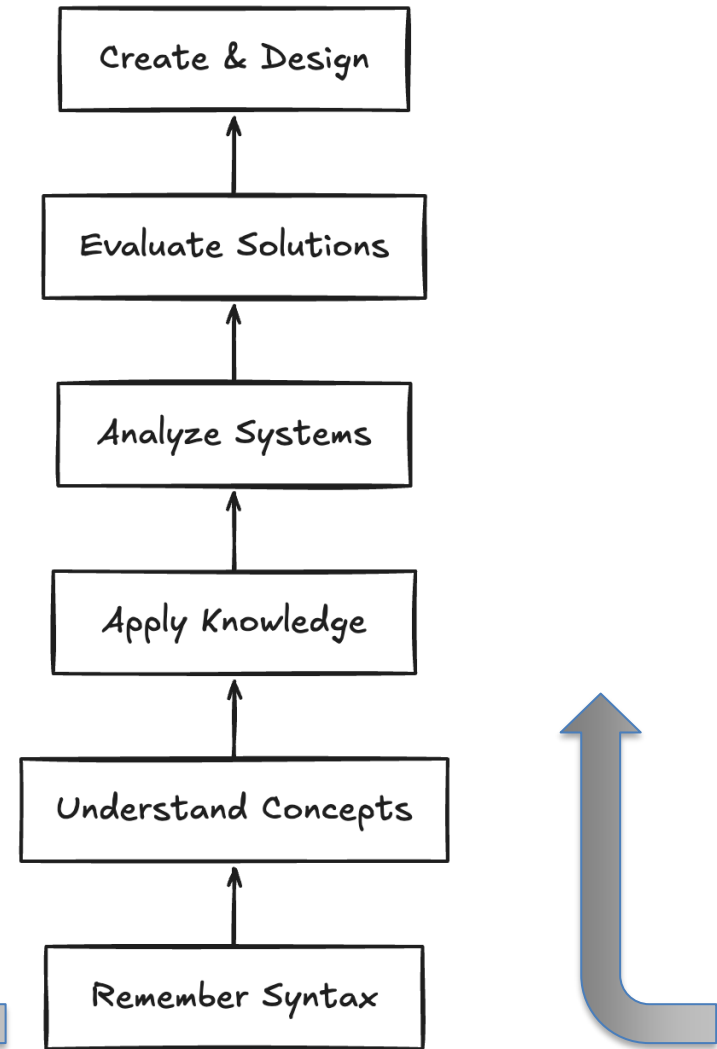


Bloom's Taxonomy Shift

- Less time on syntax
- More time on analysis and design
- AI can elevate classroom thinking

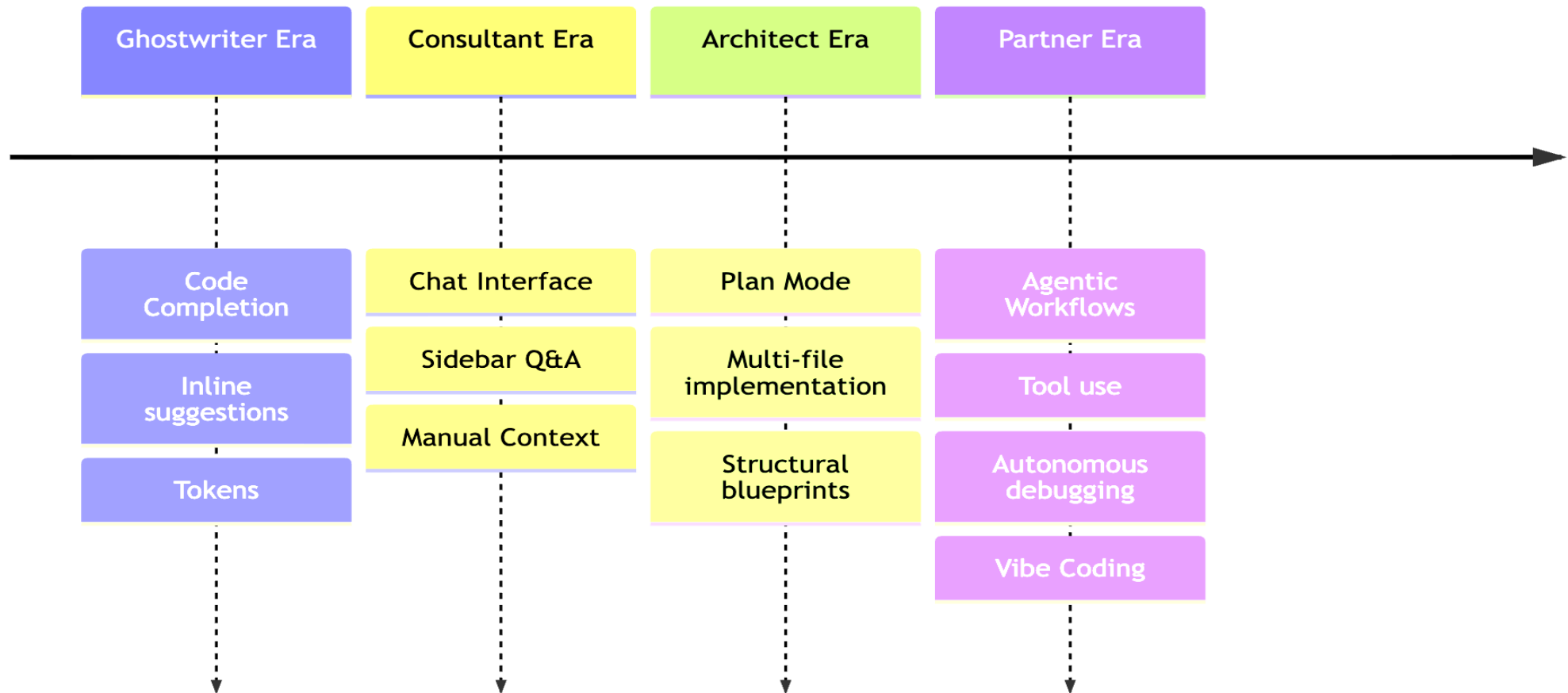
SAMR Model (Enhancement-Transformation)

- Substitution
- Augmentation
- Modification
- Redefinition



Tooling transformation

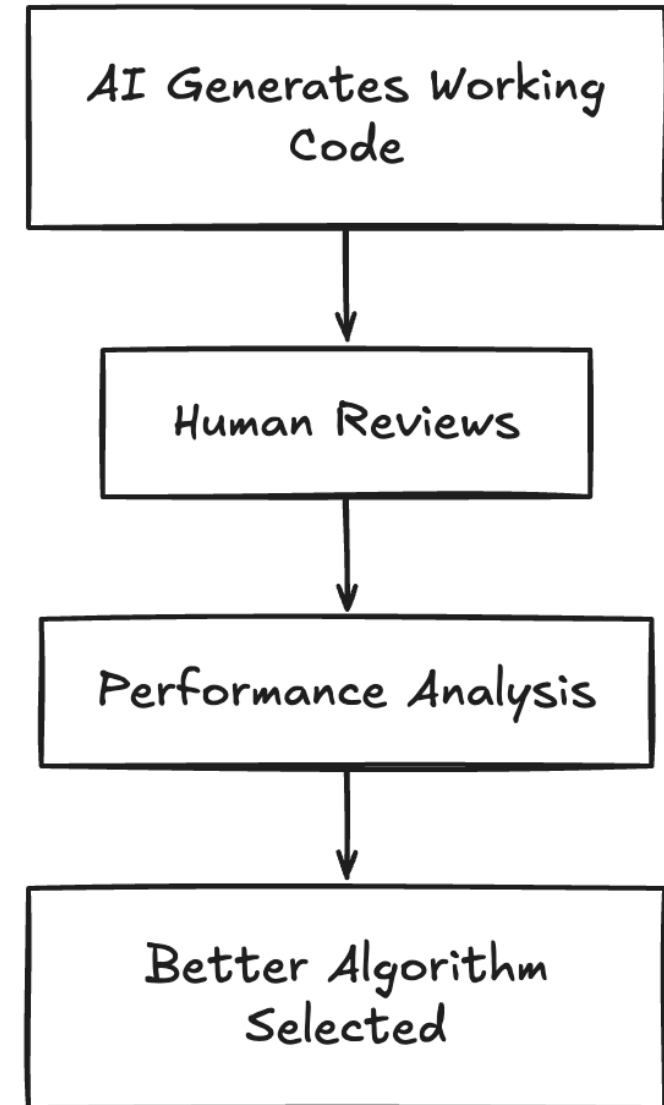
Evolution of AI Coding



Prime Numbers Example

Write a script to calculate primes less than N

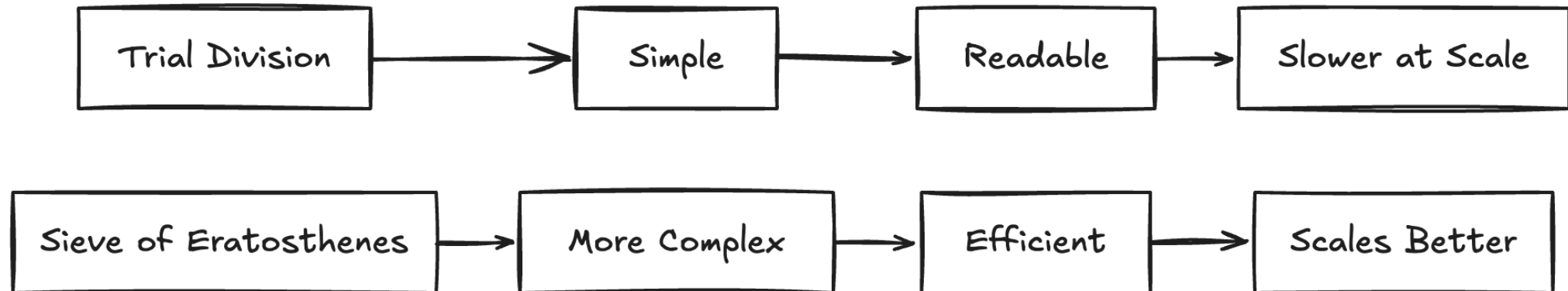
- AI often suggests simpler solutions first
- Humans improve scalability
- Correct $\neq$ optimal



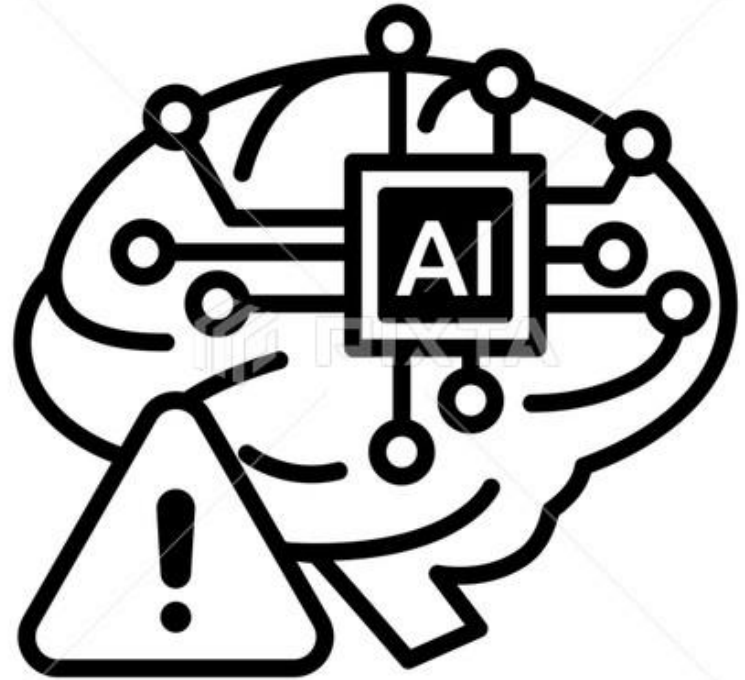
Trial Division

- Readable and beginner friendly
- Slows dramatically for large inputs

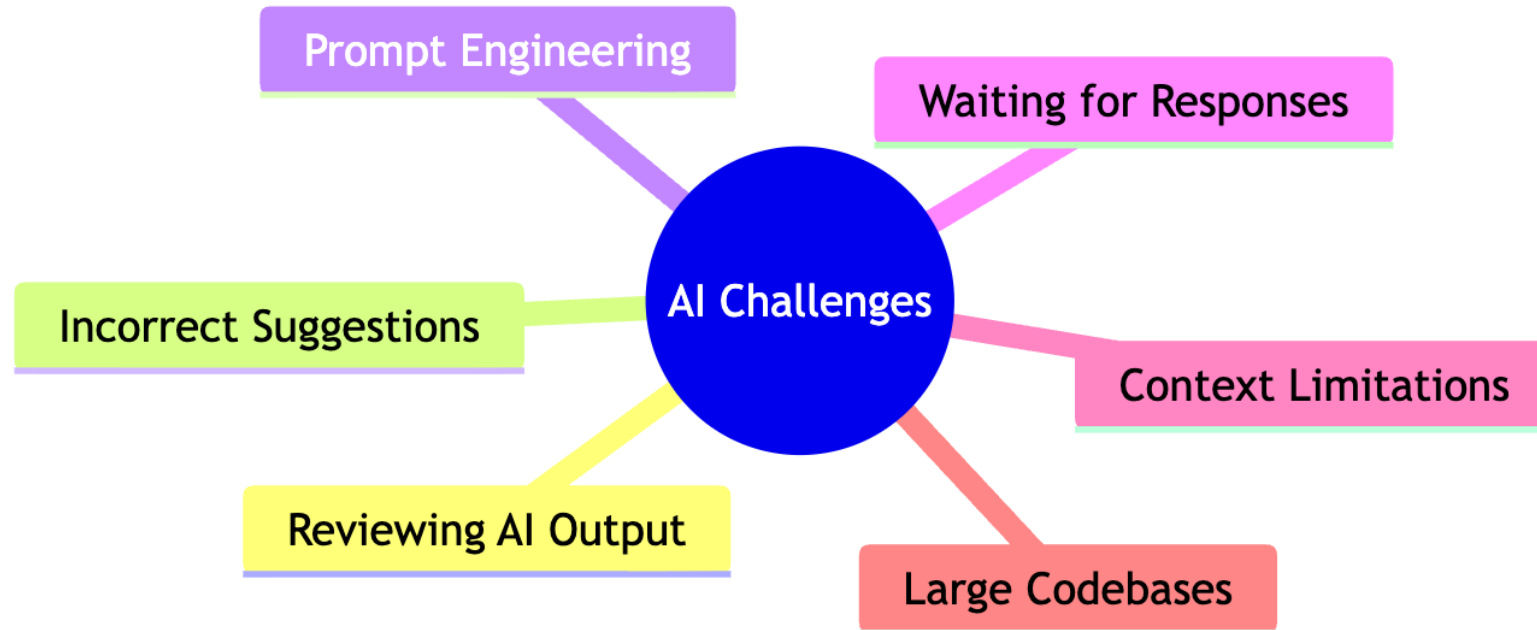
```
def is_prime(num: int) -> bool:
    """Return True if num is prime, using trial division."""
    if num < 2:
        return False
    if num == 2:
        return True
    if num % 2 == 0:
        return False
```



Risks



Research Findings (METR Study)



<https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/>

Risk mitigation plan for AI-assisted Coding in Education

Key Concerns

Blind trust

Prevent shortcut learning

Demystify the tool



Safeguards

Verify AI code & understand it before use

Use manual coding & AI to build foundational skills

Teach how AI coding tools work

Risk mitigation plan for AI-assisted Coding in Education

Key Concerns

Skills gap

No human-in-the loop



Safeguards

Foster critical thinking & peer reviews to reduce overconfidence in unverified outputs

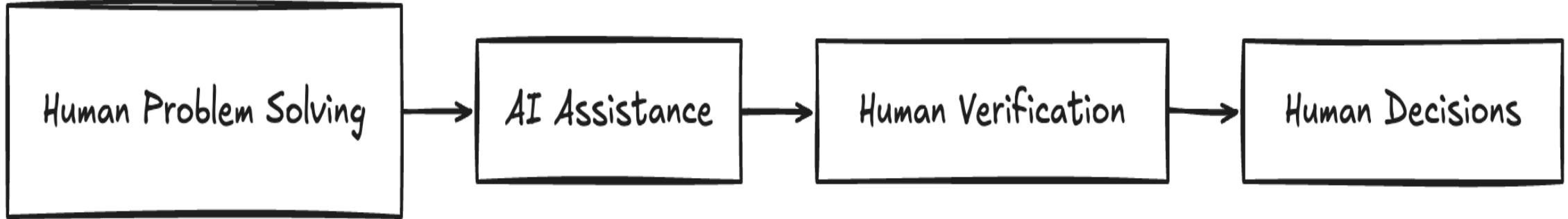
Mandate code reviews & discussions on AI-generated code

Development with AI



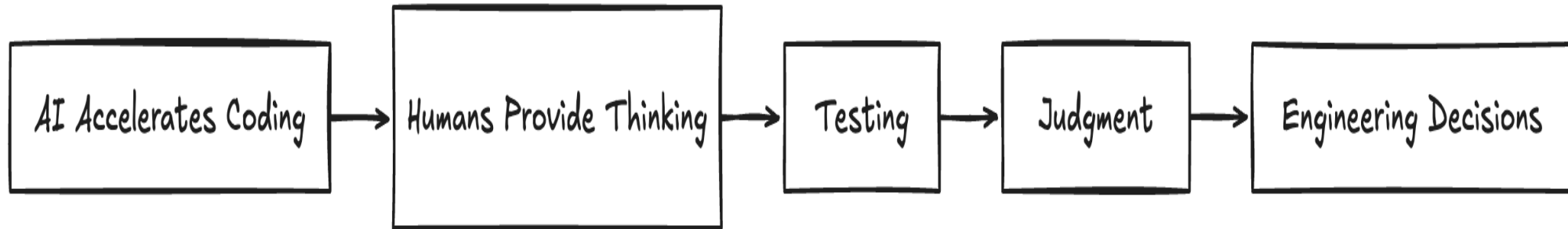
- AI generates starter code
- Students review and modify
- Human judgment remains essential

Human Thinking vs AI Suggestions



- AI accelerates implementation
- Humans optimize and evaluate
- Engineering requires tradeoffs

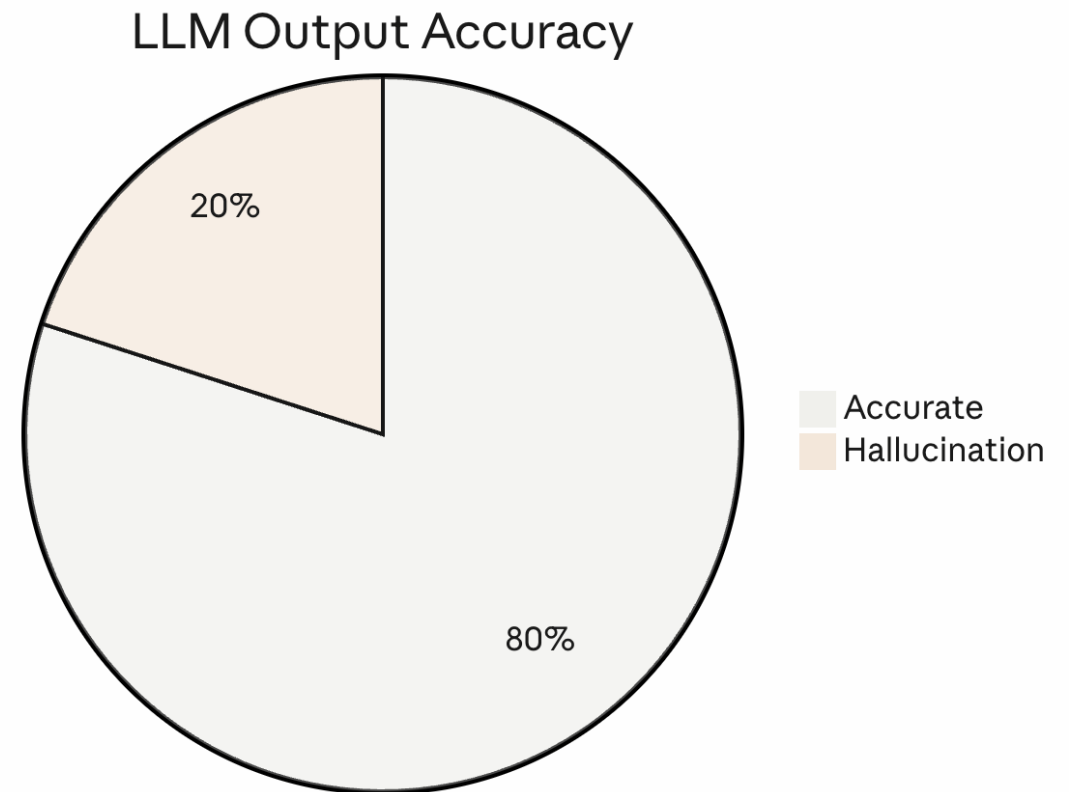
Key Message for Educators



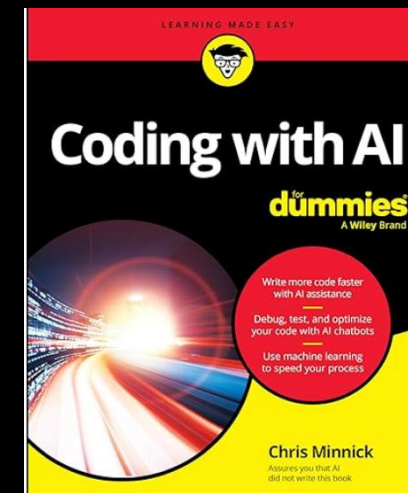
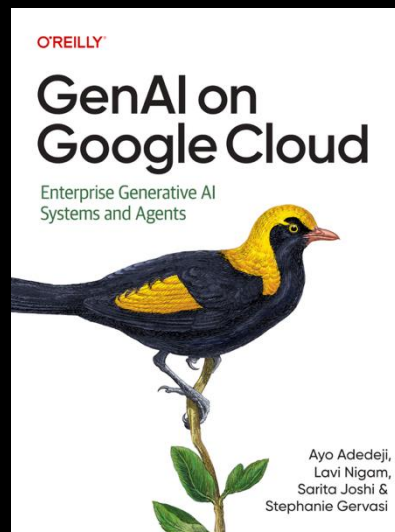
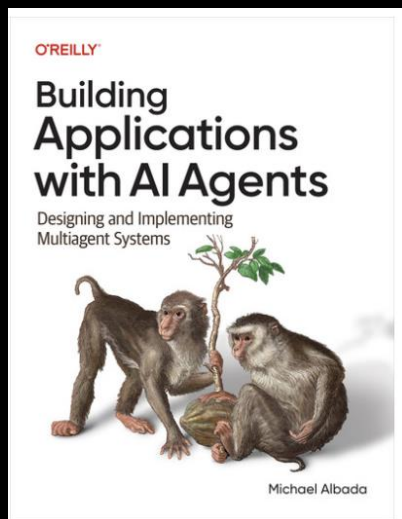
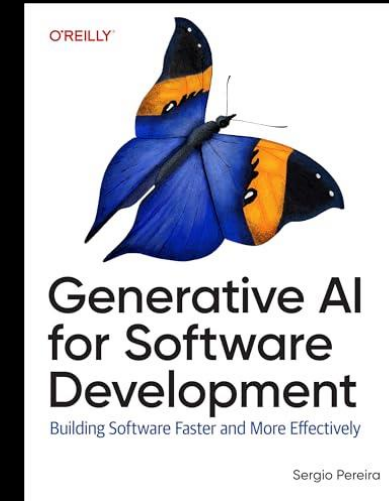
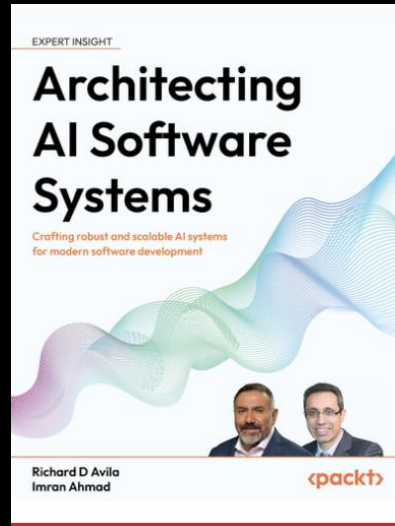
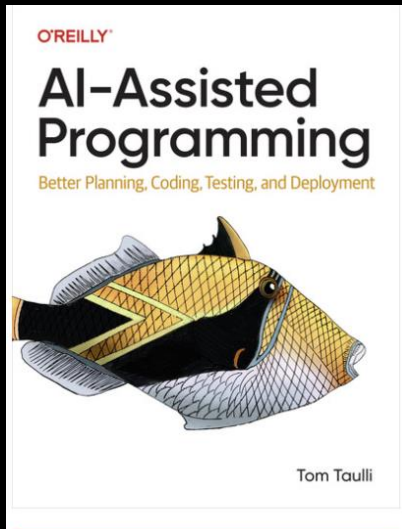
- AI does not replace thinking
- AI reduces mechanical friction
- Teach students to think critically with AI

Discussion Questions

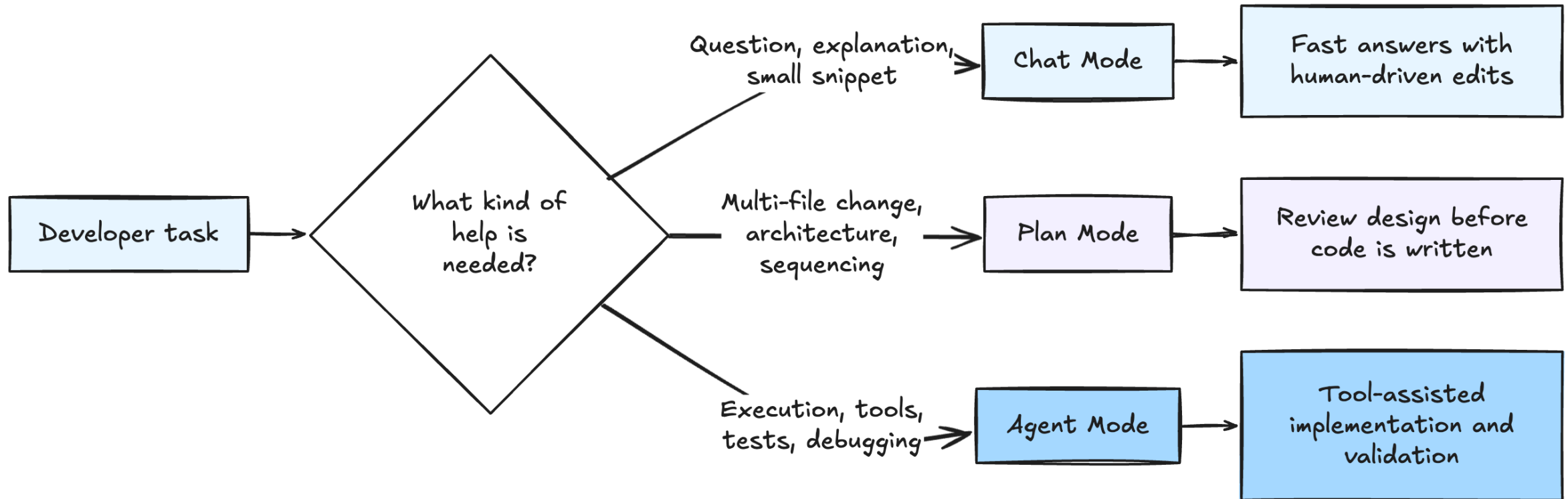
- What should AI automate?
- What thinking should humans always do?
- How do we teach responsible AI use?



Resources



IDE* Modes for AI-Assisted Coding

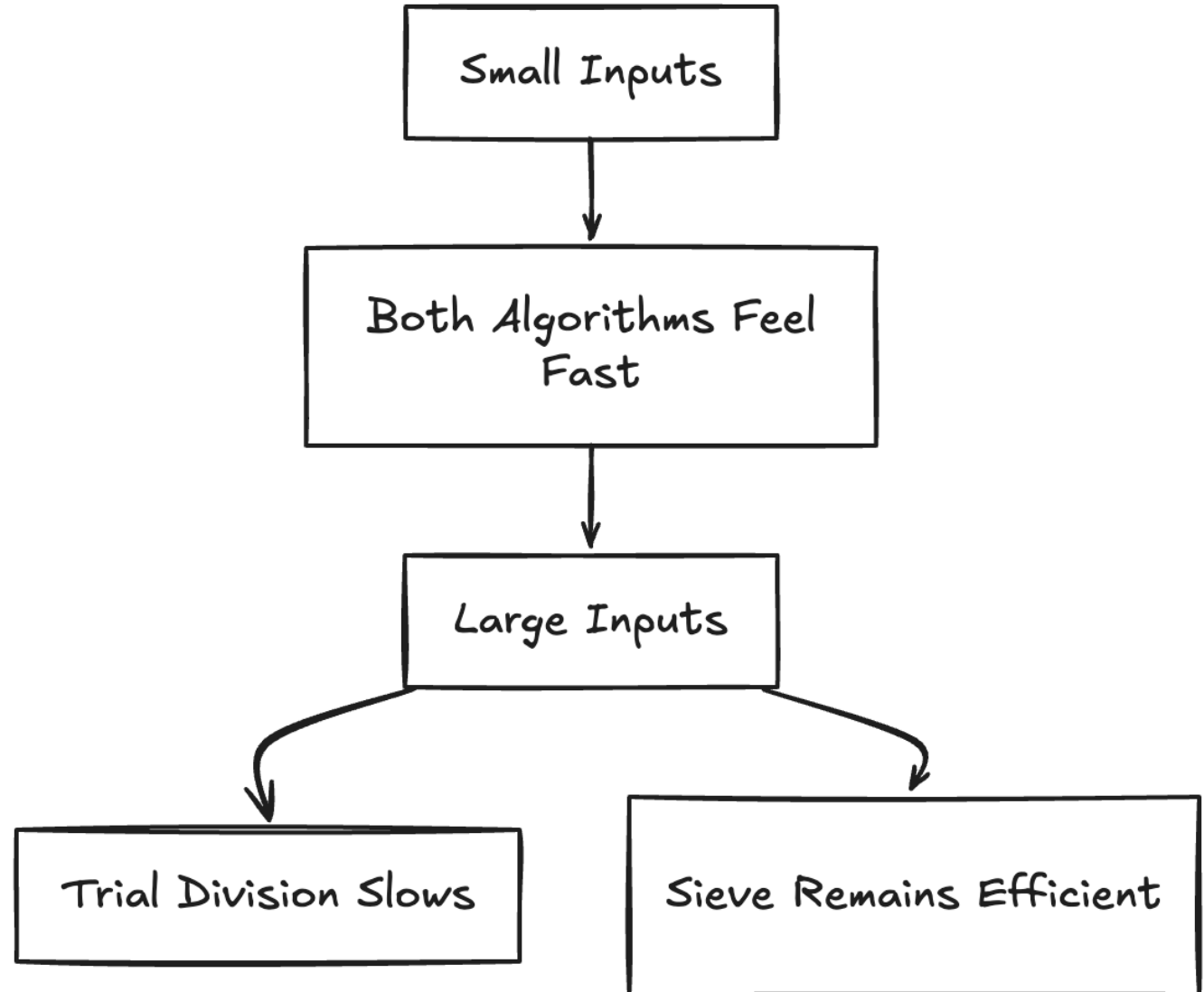


*Integrated Development Environment (IDE) – e.g.: Visual Studio Code

Sieve of Eratosthenes

- More advanced algorithm
- Time complexity: $O(n \log \log n)$
- Much faster for large inputs

```
def primes_below(n: int) -> list[int]:  
    """Return a list of all primes strictly less than n."""  
    if n < 3:  
        return []  
  
    # is_prime[i] will be False once i is known to be composite.  
    is_prime = [True] * n  
    is_prime[0] = is_prime[1] = False  
  
    for num in range(2, math.isqrt(n - 1) + 1):  
        if is_prime[num]:  
            # Mark all multiples of num (starting at num*num) as composite.  
            for multiple in range(num * num, n, num):  
                is_prime[multiple] = False  
  
    return [num for num, prime in enumerate(is_prime) if prime]
```



Research Findings (METR Study)

- Experienced developers used AI
- Tasks sometimes took 19% longer
- AI helps, but oversight matters

